UNIT 8 · TOPIC 1(A)

Ferrous Materials, Alloy Steels and Their Production

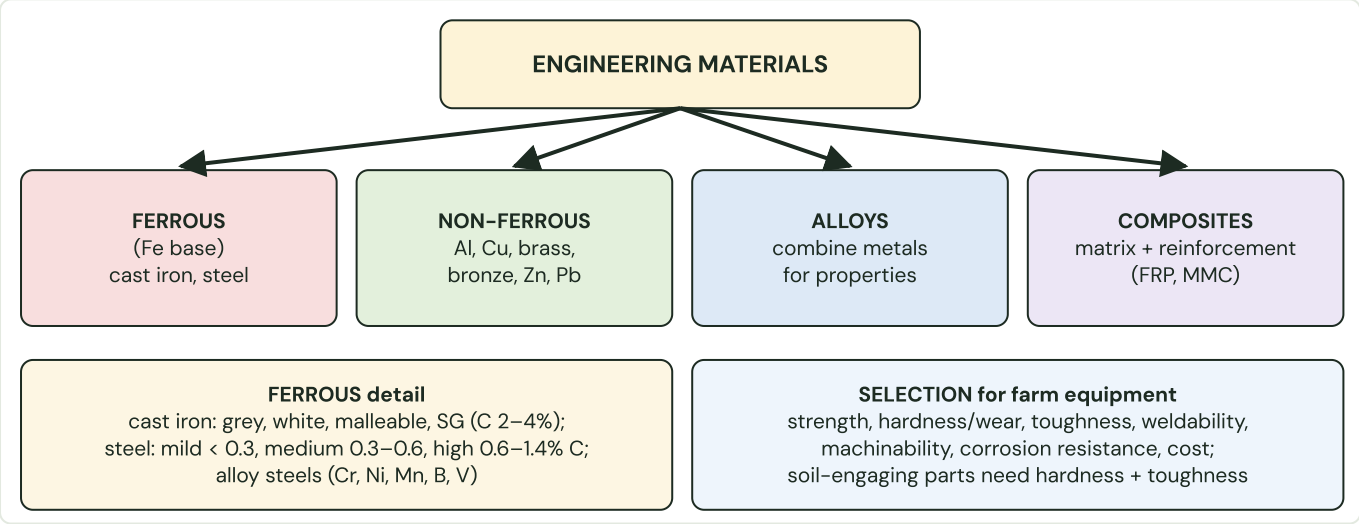
Syllabus topic: Ferrous, non-ferrous, alloy and composite materials: composition, production techniques and selection for farm equipment

Why materials matter for farm equipment

Farm machines face ABRASIVE soil wear, IMPACT/shock loads, corrosion (moisture, fertiliser, chemicals), fatigue and adhesion of sticky soil — all at low cost for price-sensitive farmers. Selection balances:

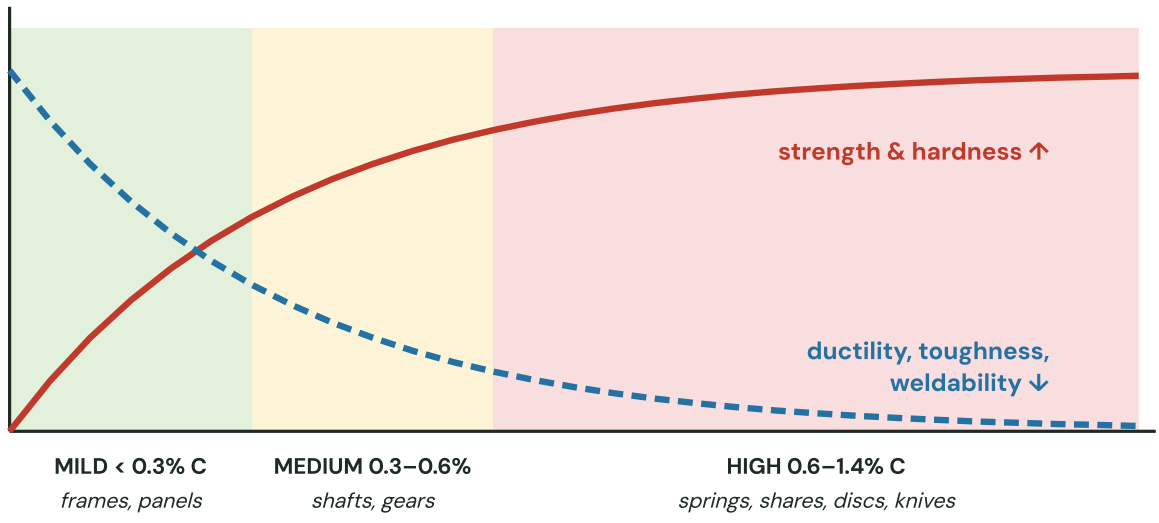
- STRENGTH, HARDNESS/wear resistance, TOUGHNESS (impact).
- Weldability, machinability, corrosion resistance, weight.
- Availability and COST.

> Soil-engaging parts need hardness AND toughness together — a hard but brittle edge chips on stones.



Engineering materials and selection criteria (redrawn from the source figure)

Plain-carbon steels (Fe + C, C up to ~1.4%)



Effect of carbon on plain-carbon steel — qualitative (redrawn; the source chart used arbitrary numeric scales)

As carbon rises, STRENGTH and HARDNESS rise while DUCTILITY, toughness and weldability fall.

Steel	Carbon	Properties	Farm uses
LOW / MILD	< 0.3%	Soft, ductile, weldable	Frames, tanks, body panels, bolts
MEDIUM-carbon	0.3–0.6%	Stronger, heat-treatable	Shafts, axles, gears, tine bars
HIGH-carbon	0.6–1.4%	Hard, wear-resistant, but brittle	Springs, cutting edges, plough shares, disc blades, knives

Small amounts of Mn and Si act as deoxidisers/strengtheners; the impurities S and P are kept low.

Alloy steels

Element	Effect
CHROMIUM	Hardness, wear, corrosion resistance — stainless if ≥ ~11% Cr
NICKEL	Toughness
MANGANESE	Hardness, work-hardening — Hadfield Mn steel for high-impact wear
MOLYBDENUM / VANADIUM	High-temperature strength, grain refinement
BORON	A dramatic rise in HARDENABILITY at tiny additions (~0.0005–0.003%) — the modern tillage-tool steel

Alloy steels are used for crankshafts, gears and high-stress and wear parts.

Cast irons (Fe + C 2–4% + Si)

Cast iron	Structure / property	Uses
GREY	Graphite flakes; excellent machinability, damping and compressive strength; cheap	Engine blocks, housings, pulleys, flywheels
WHITE	Carbon as hard cementite; very hard and brittle, wear-resistant	Chilled surfaces, grinding parts
MALLEABLE	White CI annealed for toughness	Small fittings, brackets
SPHEROIDAL-GRAPHITE (SG) / ductile / nodular	Graphite as nodules (Mg-treated) — CI castability with steel-like ductility	Crankshafts, gears, hitch parts

Production of ferrous metals

1. Iron ore → PIG IRON in a BLAST FURNACE (with coke + limestone).
2. Pig iron/scrap → STEEL by refining out carbon in a Basic Oxygen Furnace (LD converter) or an ELECTRIC ARC FURNACE (from scrap).
3. Continuous casting → rolling/forging.

Cast irons are melted in a CUPOLA or induction furnace and CAST to shape.

Steels for farm implements (as given in the source notes, July 2026)

- **BORON STEEL** is now the standard for tillage tools — mouldboard ploughs, disc harrows, cultivators, rotary blades. Tiny boron additions (~0.0005%+) sharply raise HARDENABILITY, cutting hardening cost while giving high hardness AND toughness.
- **Example grade 10B36:** ≈ 0.32–0.37% C, 1.2–1.5% Mn, ~0.0005%+ B (SSAB Boron, thyssenkrupp TBL, etc.).
- **Hardness gap (source):** plain HIGH-carbon steel (~0.70% C) after heat treatment reaches only ~38–45 HRC, but shares/points need **50–60 HRC** — so MEDIUM-carbon (~0.30% C) steel micro-alloyed with boron/chromium is preferred for the hardness + wear + toughness combination.

Metallurgy check: the maximum as-quenched hardness is set mainly by carbon content — roughly 50–55 HRC at 0.3–0.35% C and above 60 HRC at 0.7% C. Boron's contribution is hardenability: hardening through the full section rather than only at the surface. Treat the "38–45 HRC for high-carbon steel" figure as specific to the tempering practice the source cites.

Practice MCQs

Q1. Ferrous materials are based on:

(a) Aluminium (b) Iron
(c) Copper (d) Zinc

Q2. As carbon content of plain-carbon steel increases:

(a) Hardness rises, ductility falls (b) Both rise
(c) Both fall (d) No change

Q3. Mild (low-carbon) steel has carbon content:

(a) <0.3% (b) 0.3–0.6%
(c) 0.6–1.0% (d) 2–4%

Q4. High-carbon steel (0.6–1.4% C) is typically used for:

(a) Body panels
(b) Plough shares, springs, cutting edges
(c) Radiators
(d) Wiring

Q5. Cast iron contains carbon in the range:

(a) 0.1–0.3%
(b) 0.3–0.6%
(c) 2–4%
(d) 6–8%

Q6. Grey cast iron is valued for:

(a) High ductility
(b) Machinability, damping, compressive strength
(c) Weldability
(d) Corrosion resistance

Q7. The hardest and most brittle cast iron is:

(a) Grey
(b) White
(c) Malleable
(d) SG/ductile

Q8. Spheroidal-graphite (ductile) cast iron is made by treating the melt with:

(a) Sulphur
(b) Magnesium
(c) Lead
(d) Zinc

Q9. Which alloying element most increases HARDENABILITY at tiny additions?

(a) Boron
(b) Lead
(c) Tin
(d) Zinc

Q10. Chromium above about 11% makes steel:

(a) Cast iron
(b) Stainless (corrosion-resistant)
(c) Brittle only
(d) Non-magnetic wire

Q11. Manganese (Hadfield) steel is chosen for:

(a) Electrical use
(b) High-impact abrasive wear (work-hardening)
(c) Light weight
(d) Radiators

Q12. Nickel is added to steel mainly to improve:

(a) Toughness
(b) Density
(c) Colour
(d) Conductivity

Q13. Pig iron is produced in a:

(a) Cupola
(b) Blast furnace
(c) Arc furnace
(d) Ladle

Q14. Steel is made from pig iron/scrap in a:

(a) Blast furnace
(b) Basic Oxygen or Electric Arc Furnace
(c) Cupola
(d) Kiln

Q15. Cast iron for foundry work is commonly melted in a:

(a) Blast furnace
(b) Cupola
(c) Reverberatory only
(d) Rotary kiln

Q29. Boron steel grade 10B36 contains roughly:

(a) 0.32–0.37% C with ~0.0005%+ boron
(b) 2% C
(c) 12% Cr
(d) 30% Ni

Q30. Abrasion-resistant steel AR400 has a Brinell hardness of about:

(a) 100
(b) 200
(c) 360–440
(d) 700

Answers

Q1 — (b) Iron

Q2 — (a) Hardness rises, ductility falls — *Carbon raises hardness/strength but lowers ductility and weldability.*
Q3 — (a) <0.3%
Q4 — (b) Plough shares, springs, cutting edges — *Hence high-C steel for shares/springs, mild steel for panels.*
Q5 — (c) 2–4%
Q6 — (b) Machinability, damping, compressive strength — *Grey CI — machinable, damping.*
Q7 — (b) White — *White CI — hard and brittle (cementite).*
Q8 — (b) Magnesium — *SG/ductile CI — Mg-treated nodules → tough.*
Q9 — (a) Boron — *Boron gives huge hardenability at ~0.0005%+.*
Q10 — (b) Stainless (corrosion-resistant)
Q11 — (b) High-impact abrasive wear (work-hardening)
Q12 — (a) Toughness
Q13 — (b) Blast furnace
Q14 — (b) Basic Oxygen or Electric Arc Furnace
Q15 — (b) Cupola
Q29 — (a) 0.32–0.37% C with ~0.0005%+ boron — *10B36 ≈ 0.32–0.37% C with boron — as given in the source notes.*
Q30 — (c) 360–440 — *AR grades scale by BHN — AR400 ≈ 360–440.*

Sources: Standard materials-science and manufacturing texts (Callister; Kalpakjian; Rajput — Engineering Materials); BIS/IS material specifications; SSAB/industry data on tillage-tool steels (as given in the source notes, July 2026); Kepner Ch. 3 & 7 (materials for tillage tools); NPTEL materials lectures.